Some Taylor / Maclaurin Series Formulas

$$\boldsymbol{f(x)} = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

When $a = 0$, it is Maclaurin series:

$$\boldsymbol{f(x)} = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

$$\boldsymbol{e^x} = \sum_{n=0}^{\infty} \frac{1}{n!}x^n = 1 + x + \frac{1}{2!}x^2 + \cdots, \quad x \in (-\infty, +\infty)$$

$$\boldsymbol{\sin x} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}x^{2n+1} = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 + \cdots, \quad x \in (-\infty, +\infty)$$

$$\boldsymbol{\cos x} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}x^{2n} = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 + \cdots, \quad x \in (-\infty, +\infty)$$

$$\boldsymbol{\ln(1+x)} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1}x^{n+1} = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \cdots, \quad x \in (-1, 1]$$

$$\frac{\boldsymbol{1}}{\boldsymbol{1-x}} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots, \quad x \in (-1, 1)$$

$$\frac{\boldsymbol{1}}{\boldsymbol{1+x}} = \sum_{n=0}^{\infty} (-1)^n x^n = 1 - x + x^2 - x^3 + \cdots, \quad x \in (-1, 1)$$

$$\boldsymbol{(1+x)^{\alpha}} = 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}x^n = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots, \quad x \in (-1, 1)$$

$$\boldsymbol{\arctan x} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}x^{2n+1} = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 + \cdots, \quad x \in [-1, 1]$$

$$\boldsymbol{\arcsin x} = \sum_{n=0}^{\infty} \frac{(2n)!}{4^n(n!)^2(2n+1)}x^{2n+1} = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \frac{5}{112}x^7 + \frac{35}{1152}x^9 + \cdots, \quad x \in (-1, 1)$$

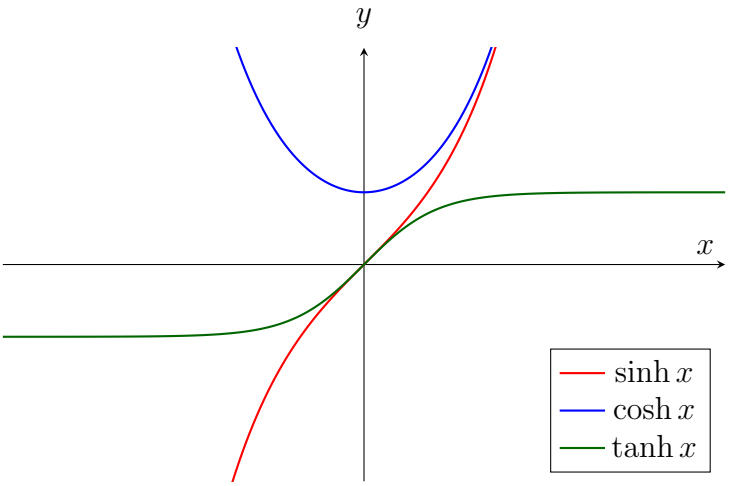
$$\boldsymbol{\tan x} = \sum_{n=1}^{\infty} \frac{B_{2n}(-4)^n\,(1-4^n)}{(2n)!}x^{2n-1} = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \frac{17}{315}x^7 + \frac{62}{2835}x^9 + \cdots, \quad x \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

Hyperbolic Trigonometric Identities

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

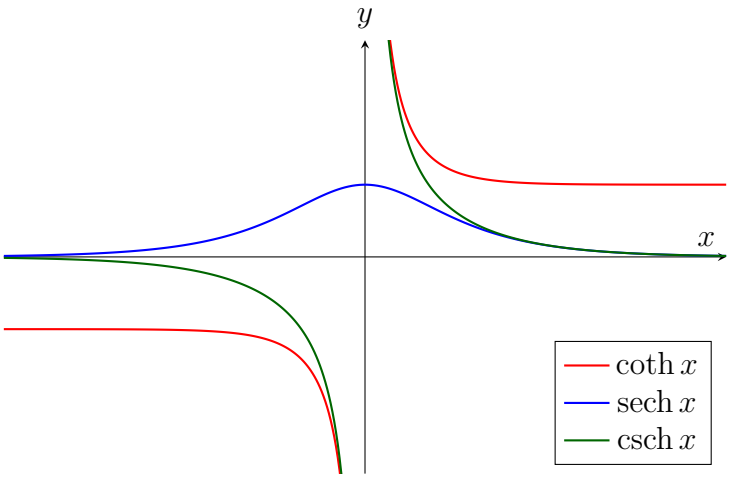
$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}} \qquad \text{for } x \neq 0$$

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

$$\operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}} \qquad \text{for } x \neq 0$$



Trigonometric Identities

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$\sin^2 \frac{x}{2} = \frac{1 - \cos x}{2}$$

$$\cos^2 \frac{x}{2} = \frac{1 + \cos x}{2}$$

$$\tan \frac{x}{2} = \frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}$$

$$\sin A + \sin B = 2 \sin \frac{A + B}{2} \cos \frac{A - B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A + B}{2} \sin \frac{A - B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A + B}{2} \cos \frac{A - B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A + B}{2} \sin \frac{A - B}{2}$$

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$\sec x = \frac{1}{\cos x}, \qquad \csc x = \frac{1}{\sin x}, \qquad \cot x = \frac{\cos x}{\sin x}$$

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x, \qquad \cos\left(\frac{\pi}{2} - x\right) = \sin x, \qquad \tan\left(\frac{\pi}{2} - x\right) = \cot x$$

Hyperbolic Functions

$$\cosh^2 x - \sinh^2 x = 1$$

$$\sinh 2x = 2\sinh x \cosh x$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x$$

$$\cosh^{-1} x = \ln\left(x + \sqrt{x^2 - 1}\right) \qquad (x \geq 1)$$

$$\sinh^{-1} x = \ln\left(x + \sqrt{x^2 + 1}\right)$$

$$\tanh^{-1} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) \qquad (|x| < 1)$$

$$\int \operatorname{sech}^2 x \, \mathrm{d}x = \tanh x + C$$

$$\int \operatorname{csch}^2 x \, \mathrm{d}x = -\coth x + C$$

$$\int \operatorname{sech} x \tanh x \, \mathrm{d}x = -\operatorname{sech} x + C$$

$$\int \operatorname{csch} x \coth x \, \mathrm{d}x = -\operatorname{csch} x + C$$

$$\int \frac{1}{\sqrt{x^2 + 1}} \, \mathrm{d}x = \sinh^{-1} x + C$$

$$\int \frac{1}{\sqrt{x^2 - 1}} \, \mathrm{d}x = \cosh^{-1} x + C \qquad (x > 1)$$

$$\int \frac{1}{1 - x^2} \, \mathrm{d}x = \tanh^{-1} x + C \qquad (|x| < 1)$$

Algebraic Identities

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$a^2 - 2ab + b^2 = (a - b)^2$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + 3a^2b + 3ab^2 + b^3 = (a + b)^3$$

$$a^3 - 3a^2b + 3ab^2 - b^3 = (a - b)^3$$

$$a^4 - b^4 = (a - b)(a + b)(a^2 + b^2)$$

$$a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \cdots + ab^{n-2} + b^{n-1})$$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$

$$x^2 - (a + b)x + ab = (x - a)(x - b)$$

$$\frac{1}{a} + \frac{1}{b} = \frac{a + b}{ab}$$

$$\frac{1}{a - b} - \frac{1}{a + b} = \frac{2b}{a^2 - b^2}$$

Differentiation and Integration

$$\frac{\mathrm{d}}{\mathrm{d}x}(uv) = u'v + uv'$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(f(g(x))\big) = f'(g(x))g'(x)$$

$$\int \frac{1}{a^2 + x^2}\,\mathrm{d}x = \frac{1}{a}\arctan\frac{x}{a} + C$$

$$\int u'v\,\mathrm{d}x = uv - \int uv'\,\mathrm{d}x$$

$$\int_a^b f(g(x))g'(x)\,\mathrm{d}x = \int_{g(a)}^{g(b)} f(u)\,\mathrm{d}u$$

$$\int_a^b f'(x)\,\mathrm{d}x = f(b) - f(a)$$

$$\operatorname{Var}(X) = E\big((X-\mu)^2\big) = E(X^2) - \mu^2$$

$$\operatorname{Cov}(X,Y) = E\big((X-\mu_X)(Y-\mu_Y)\big) = E(XY) - \mu_X\mu_Y$$

$$\operatorname{Var}(aX \pm bY) = a^2\operatorname{Var}(X) + b^2\operatorname{Var}(Y) \pm 2ab\,\operatorname{Cov}(X,Y)$$

$$\rho = \frac{\operatorname{Cov}(X,Y)}{\sigma_X\sigma_Y}$$

For independent random variables X and Y:

$$E(XY) = E(X)E(Y)$$

$$\operatorname{Var}(aX \pm bY) = a^2\operatorname{Var}(X) + b^2\operatorname{Var}(Y)$$

Common Cumulative Distribution Functions (CDFs)

The CDF of a random variable X is $F_X(x) = P(X \leq x)$.
The PDF of a random variable X is $f_X(x) = F'_X(x)$.
When $X \sim N(\mu, \sigma^2)$:

$$F_X(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{x-\mu}{\sigma}} e^{-t^2/2}\,\mathrm{d}t \qquad f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

When $X \sim \mathrm{Bin}(n, p)$:

$$F_X(x) = \sum_{r=0}^{\lfloor x \rfloor} \binom{n}{r} p^r (1-p)^{n-r} \qquad P_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

When $X \sim \mathrm{Geo}(p)$:

$$F_X(x) = \sum_{r=1}^{\lfloor x \rfloor} p(1-p)^{r-1} \qquad P_X(x) = p(1-p)^{x-1}$$

When $X \sim \mathrm{Po}(\lambda)$:

$$F_X(x) = e^{-\lambda} \sum_{r=0}^{\lfloor x \rfloor} \frac{\lambda^r}{r!} \qquad P_X(x) = e^{-\lambda} \frac{\lambda^x}{x!}$$

e

$$\begin{aligned} e &= \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \\ &= 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots \end{aligned}$$